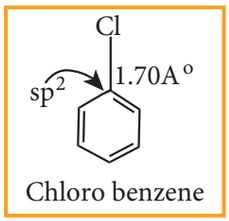
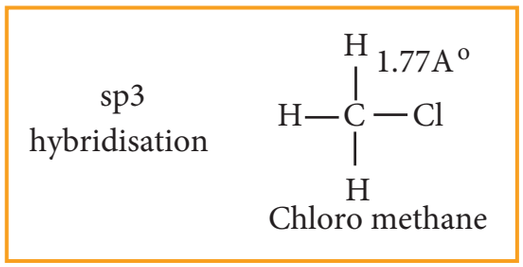
shorter in length and stronger than in halo alkanes.

Example



Chloro benzene

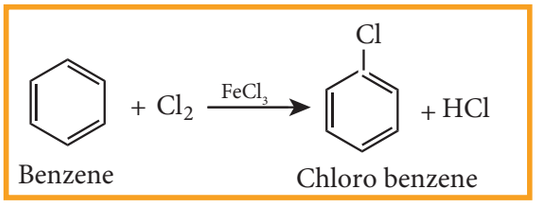


Chloro methane

14.5.3 Methods of preparation

1) Direct halogenation

Chlorobenzene is prepared by the direct chlorination of benzene in the presence of lewis acid catalyst like $FeCl_3$



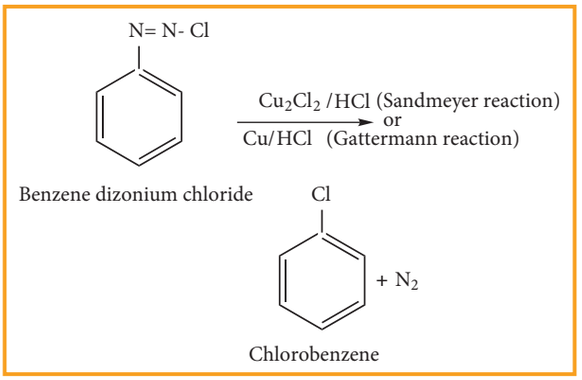
Benzene Chloro benzene

2) From benzene diazonium chloride

Chloro benzene is prepared by Sandmeyer reaction or Gattermann reaction using benzene diazonium chloride.

(i) Sandmeyer reaction

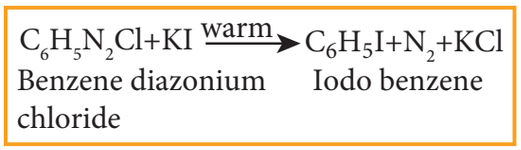
When aqueous solution of benzene diazonium chloride is warmed with Cu_2Cl_2 in HCl gives chloro benzene



Chlorobenzene

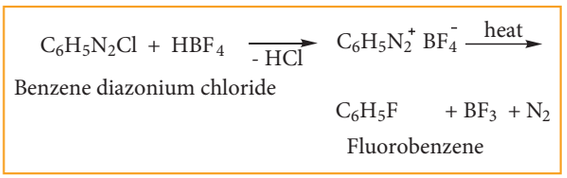
3) Preparation of iodobenzene

Iodobenzene is prepared by warming benzene diazonium chloride with aqueous KI solution.



4) Preparation of fluorobenzene

Fluoro benzene is prepared by treating benzenediazonium chloride with fluoro boric acid. This reaction produces diazonium fluoroborate which on heating produces fluorobenzene. This reaction is called Balz – schiemann reaction.



5) Commercial preparation of chloro benzene (Raschig process)

Chloro benzene is commercially prepared by passing a mixture of benzene vapour, air and HCl over heated cupric chloride .This reaction is called Raschig process.

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